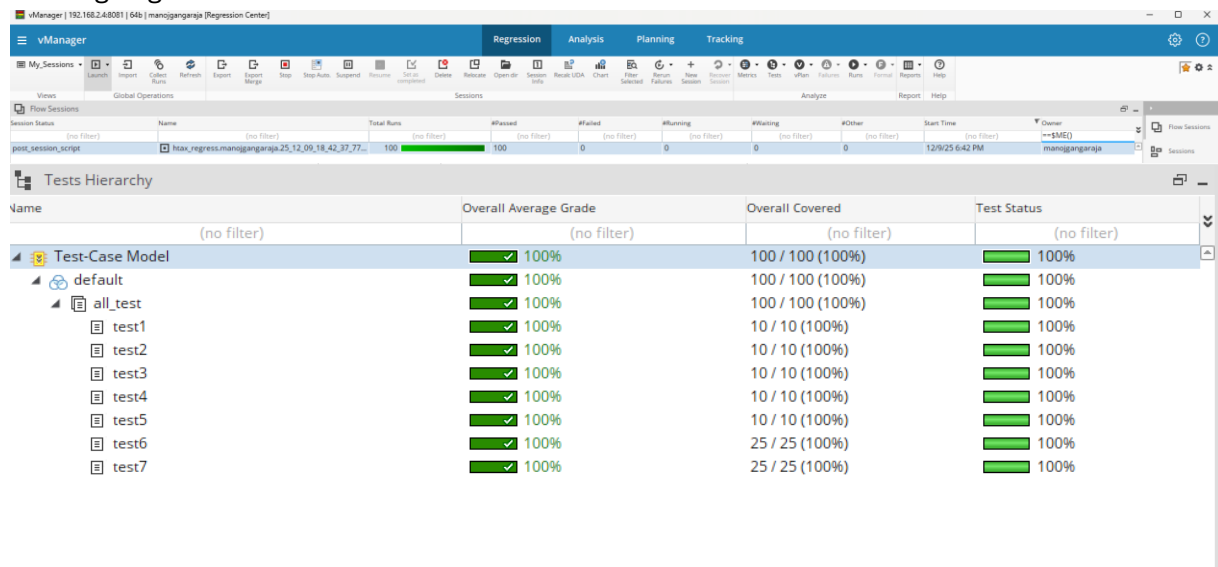


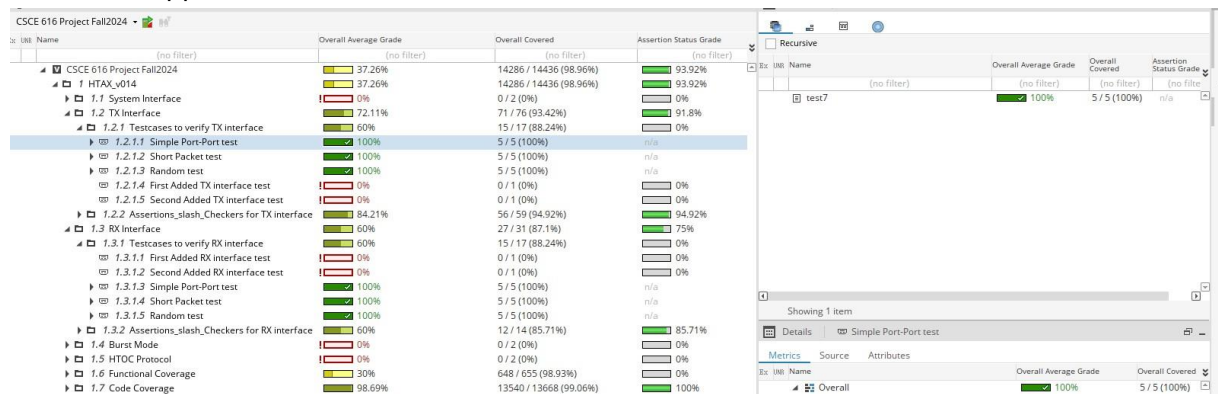
CSCE Lab 10 Report:

Coverage report:

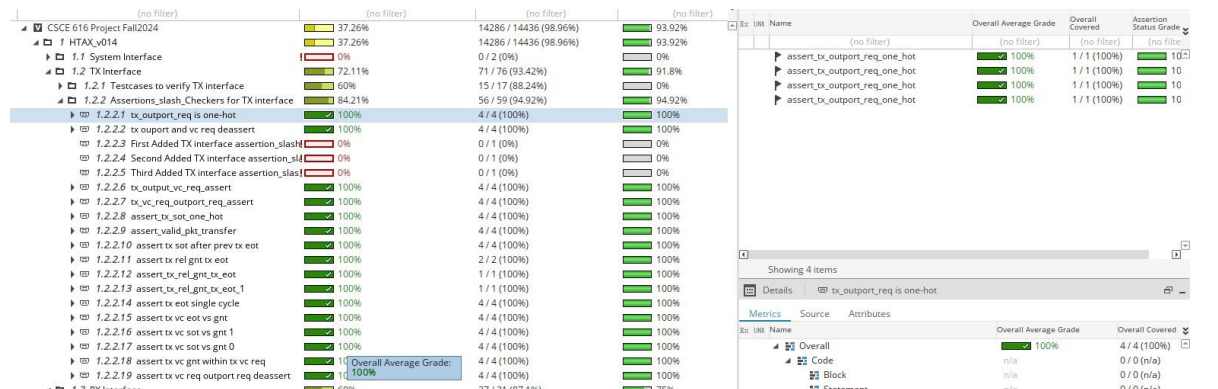
Passing Regression after fix:



Testcase mapped:



Assertion/Checkers for TX:



Assertion/Checkers for RX:

CSCE 616 Project Fall2024	37.26%	14286 / 14436 (98.96%)	93.92%
HTAX_v014	37.26%	14286 / 14436 (98.96%)	93.92%
1.1 System Interface	0%	0 / 2 (0%)	0%
1.2 TX Interface	72.11%	71 / 76 (93.42%)	91.8%
1.2.1 Testcases to verify TX interface	60%	15 / 17 (88.24%)	0%
1.2.2 Assertions_slash_Checkers for TX interface	84.21%	56 / 59 (94.92%)	94.92%
1.3 RX Interface	60%	27 / 31 (87.1%)	75%
1.3.1 Testcases to verify RX interface	60%	15 / 17 (88.24%)	0%
1.3.2 Assertions_slash_Checkers for RX interface	60%	12 / 14 (85.71%)	85.71%
1.3.2.1 First Added RX interface assertion_slash	0%	0 / 1 (0%)	0%
1.3.2.2 Second Added RX interface assertion_slash	0%	0 / 1 (0%)	0%
1.3.2.3 assert_eot_timeout_check	100%	4 / 4 (100%)	100%
1.3.2.4 assert_rx_sot_one_hot	100%	4 / 4 (100%)	100%
1.3.2.5 assert_rx_eot_single_cycle	100%	4 / 4 (100%)	100%
1.4 Burst Mode	0%	0 / 2 (0%)	0%

(no filter)	(no filter)	(no filter)	(no filter)
assert_eot_timeout_check	100%	1 / 1 (100%)	10
assert_eot_timeout_check	100%	1 / 1 (100%)	10
assert_eot_timeout_check	100%	1 / 1 (100%)	10
assert_eot_timeout_check	100%	1 / 1 (100%)	10

Functional Coverage for TX:

CSCE 616 Project Fall2024	37.26%	14286 / 14436 (98.96%)	93.92%
HTAX_v014	37.26%	14286 / 14436 (98.96%)	93.92%
1.1 System Interface	0%	0 / 2 (0%)	0%
1.2 TX Interface	72.11%	71 / 76 (93.42%)	91.8%
1.3 RX Interface	60%	27 / 31 (87.1%)	75%
1.4 Burst Mode	0%	0 / 2 (0%)	0%
1.5 HTOC Protocol	0%	0 / 2 (0%)	0%
1.6 Functional Coverage	30%	648 / 655 (98.93%)	0%
1.6.1 System Interface	0%	0 / 1 (0%)	0%
1.6.2 TX Interface	75%	628 / 631 (99.52%)	0%
1.6.2.1 VC Request	100%	12 / 12 (100%)	n/a
1.6.2.2 Outport Request	100%	16 / 16 (100%)	n/a
1.6.2.3 Packet Data Length	0%	0 / 1 (0%)	0%
1.6.2.4 First Added TX interface coverage	0%	0 / 1 (0%)	0%
1.6.2.5 Second added TX interface coverage	0%	0 / 1 (0%)	0%
1.6.2.6 VC Gnt	100%	12 / 12 (100%)	n/a
1.6.2.7 dest port	100%	16 / 16 (100%)	n/a
1.6.2.8 vc	100%	12 / 12 (100%)	n/a
1.6.2.9 length	100%	64 / 64 (100%)	n/a
1.6.2.10 x dest port vc	100%	48 / 48 (100%)	n/a
1.6.2.11 x dest port length	100%	256 / 256 (100%)	n/a
1.6.2.12 vc length	100%	192 / 192 (100%)	n/a

(no filter)	(no filter)	Covered	Status Grade
TX_VC_REQ	100%	3 / 3 (100%)	n/a
TX_VC_REQ	100%	3 / 3 (100%)	n/a
TX_VC_REQ	100%	3 / 3 (100%)	n/a
TX_VC_REQ	100%	3 / 3 (100%)	n/a

Showing 4 items

Details VC Request

Metrics Source Attributes

Ex Name Overall Average Grade Overall Covered

Functional Coverage for RX:

CSCE 616 Project Fall2024	37.26%	14286 / 14436 (98.96%)	93.92%
HTAX_v014	37.26%	14286 / 14436 (98.96%)	93.92%
1.1 System Interface	0%	0 / 2 (0%)	0%
1.2 TX Interface	72.11%	71 / 76 (93.42%)	91.8%
1.3 RX Interface	60%	27 / 31 (87.1%)	75%
1.4 Burst Mode	0%	0 / 2 (0%)	0%
1.5 HTOC Protocol	0%	0 / 2 (0%)	0%
1.6 Functional Coverage	30%	648 / 655 (98.93%)	0%
1.6.1 System Interface	0%	0 / 1 (0%)	0%
1.6.2 TX Interface	75%	628 / 631 (99.52%)	0%
1.6.3 RX Interface	75%	20 / 21 (95.24%)	0%
1.6.3.1 First Added RX interface coverage	0%	0 / 1 (0%)	0%
1.6.3.2 rx_eot	100%	8 / 8 (100%)	n/a
1.6.3.3 rx_data	100%	4 / 4 (100%)	n/a
1.6.3.4 rx_s Metrics Port: rx_eot	100%	8 / 8 (100%)	n/a
1.6.4 Burst Mode	0%	0 / 1 (0%)	0%

(no filter)	(no filter)	Covered	Status Grade
RX_EOT	100%	2 / 2 (100%)	n/a
RX_EOT	100%	2 / 2 (100%)	n/a
RX_EOT	100%	2 / 2 (100%)	n/a
RX_EOT	100%	2 / 2 (100%)	n/a

Code Coverage:

CSCE 616 Project Fall2024	37.26%	14286 / 14436 (98.96%)	93.92%
HTAX_v014	37.26%	14286 / 14436 (98.96%)	93.92%
1.1 System Interface	0%	0 / 2 (0%)	0%
1.2 TX Interface	72.11%	71 / 76 (93.42%)	91.8%
1.3 RX Interface	60%	27 / 31 (87.1%)	75%
1.4 Burst Mode	0%	0 / 2 (0%)	0%
1.5 HTOC Protocol	0%	0 / 2 (0%)	0%
1.6 Functional Coverage	30%	648 / 655 (98.93%)	0%
1.7 Code Coverage	98.69%	13540 / 13668 (99.06%)	100%
1.7.1 Block	98.69%	3385 / 3417 (99.06%)	100%
1.7.2 Expression	98.69%	3385 / 3417 (99.06%)	100%
1.7.3 Toggle	98.69%	3385 / 3417 (99.06%)	100%
1.7.4 FSM	98.69%	3385 / 3417 (99.06%)	100%

Ex	Name	Overall Average Grade	Overall Covered	Assertion Status Grade
top	(no filter)	98.69%	3385 / 3417...	100%

Bugs report:

Bug 1

What is the bug?

Behaviour:

- 1) rx eot was never observed.
- 2) The issue occurred when tx eot was asserted for all destination ports, particularly within a fork--join block. Tracing eot in through htax output data mux showed that when all tx eot signals were high, eot_in became 4'b1111. This caused selected_eot—which drives rx eot—to be forced to 0, resulting in an assertion failure at the rx interface.

Where is it?

Module: Design module

File: design/htax_output_data_mux.v Line

number(s): 43

How to reproduce:

The bug TIMEOUT rx eot did not occur within 1000 cycles after rx sot was observed during short-packet testing using: xrun -f run.f +UVM_TESTNAME=short_packet_test

Bug fix:

```
work > design > htax_output_data_mux.v
25 );
33 begin
35     case (inport_sel)
37         4'b1111: selected_sot = sot_in[((3*VC)-1):(2*VC)];
38         4'b1110: selected_sot = sot_in[((2*VC)-1):(1*VC)];
39         4'b1101: selected_sot = sot_in[((1*VC)-1):(0*VC)];
40     endcase
41 end
42
43 assign selected_eot = |(eot_in & inport_sel_reg); //& ~(eot_in));
44
45 `ifdef ASYNC_RES
46 always @(posedge clk or negedge res_n) `else
47 always @(posedge clk) `endif
48 begin
49 if (!res_n) begin
50     inport_sel_reg <= {NUM_PORTS{1'b0}};
51     any_gnt_reg <= 0;
52 end else begin
53     any_gnt_reg <= any_gnt;
```

Fixed code:

assign selected_eot = |(eot_in & inport_sel_reg);

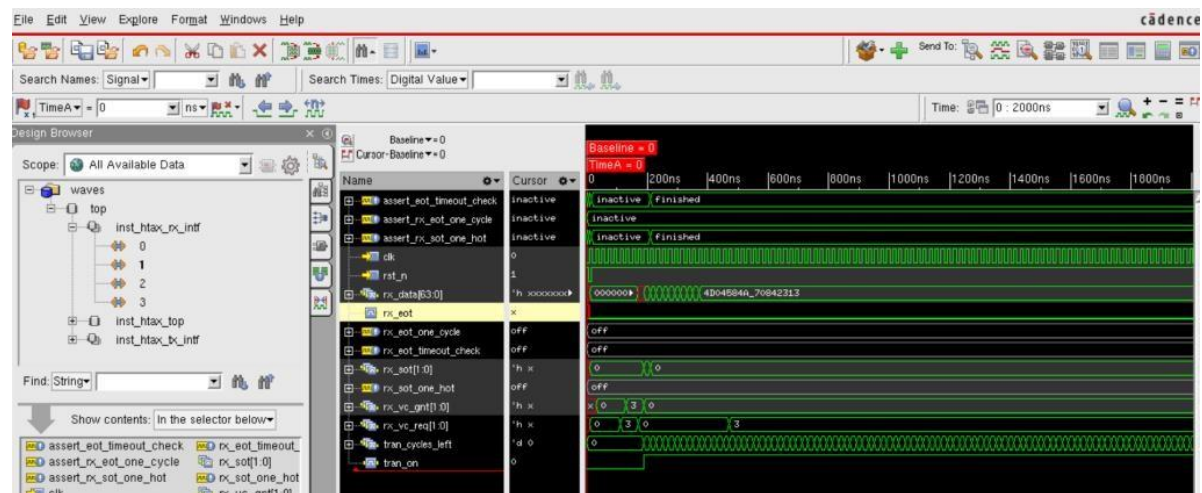
Failing Assertion:

```
vc          integral      2      'h3
length      integral      32     'ha
data        da( integral) 10     -
[0]         integral      64     'hf77baae1fa2e7e49
[1]         integral      64     'hec164e0d68b95665
[2]         integral      64     'h2b29ab9e6cafcdf1
[3]         integral      64     'hd23fcb2d58b31300
[4]         integral      64     'h64e1f50ac104ae97
[5]         integral      64     'hc6cf6c1c834b180f
[6]         integral      64     'h1d35c96f9853d4ff
[7]         integral      64     'h97a2adf970cb1163
[8]         integral      64     'h5645f137df61c037
[9]         integral      64     'h10641a4082c65a1b
begin_time  time          64     370000
depth       int           32     'd2
parent sequence (name) string    22     short_packet_vsequence
parent sequence (full name) string 49     uvm_test_top.tb.vsequencer.short_packet_vsequence
sequencer   string        36     uvm_test_top.tb.tx_port[0].sequencer
-----
xmsim: *F,ASRTST ( ../tb/htax_rx_interface.sv,56): (time 20210 NS) Assertion top.inst_htax_rx_intf[3].assert_eot_timeout_check has failed
Memory Usage - Current physical: 114.4M, Current virtual: 161.8M
CPU Usage - 0.0s system + 0.1s user = 0.1s total (8.3% cpu)
Simulation terminated via $fatal(2) at time 20210 NS + 2
../tb/htax_rx_interface.sv:56      $fatal("HTAX_RX_INF ERROR : TIMEOUT rx_eot did not occur within 1000 cycles after rx_sot");
xcelium> exit

coverage setup:
workdir : ./cov_work
dutinst : top(top)
scope   : scope
testname : test

coverage files:
model(design data) : ./cov_work/scope/icc_4e8e3c4e_7997b529.ucm (reused)
```

Failing Scenario Waveform:



Failing Assertion Passing after the fix:

```
manojgangaraja@n01-zeus:~/lab-10-manojg24/work/sim
data      da(integral)      10      -
[0]       integral         64      'h66428166d5561d4c
[1]       integral         64      'hc3c74ff4312f88dle
[2]       integral         64      'h235f26aea589d169
[3]       integral         64      'hc02c41eff7676c8e
[4]       integral         64      'h97c8b237d5b1aebb
[5]       integral         64      'h1d45b5f382618b46
[6]       integral         64      'h57e79da33c503ff7f
[7]       integral         64      'hf6c160ac2c399a0c
[8]       integral         64      'h88cf3d5e8f9af708
[9]       integral         64      'hc3abdeed52f75ec9

UVM_INFO ../tb/htax_scoreboard_c.sv(117) @ 169290000: uvm_test_top.tb.htax_sb [SCOREBOARD] Data mat
UVM_INFO ../tb/htax_scoreboard_c.sv(120) @ 169290000: uvm_test_top.tb.htax_sb [SCOREBOARD] Dropping
UVM_INFO /opt/coe/cadence/XCELIUM/tools/methodology/UVM/CDNS-1.1d/sv/src/base/uvm_objection.svh(126)
UVM_INFO ../tb/htax_scoreboard_c.sv(150) @ 219270000: uvm_test_top.tb.htax_sb [SCOREBOARD] End of S
UVM_INFO ../tb/htax_scoreboard_c.sv(152) @ 219270000: uvm_test_top.tb.htax_sb [SCOREBOARD] Port 0 0
UVM_INFO ../tb/htax_scoreboard_c.sv(156) @ 219270000: uvm_test_top.tb.htax_sb [SCOREBOARD] Port 1 0
UVM_INFO ../tb/htax_scoreboard_c.sv(160) @ 219270000: uvm_test_top.tb.htax_sb [SCOREBOARD] Port 2 0
UVM_INFO ../tb/htax_scoreboard_c.sv(164) @ 219270000: uvm_test_top.tb.htax_sb [SCOREBOARD] Port 3 0

--- UVM Report catcher Summary ---

Number of demoted UVM_FATAL reports : 0
Number of demoted UVM_ERROR reports : 0
Number of demoted UVM_WARNING reports: 0
Number of caught UVM_FATAL reports : 0
Number of caught UVM_ERROR reports : 0
Number of caught UVM_WARNING reports : 0

--- UVM Report Summary ---

** Report counts by severity
UVM_INFO : 2416
UVM_WARNING : 0
UVM_ERROR : 0
UVM_FATAL : 0
** Report counts by id
[RNIST] 1
[SCOREBOARD] 1605
[TEST_DONE] 1
[TOP] 5
[UVMTOP] 1
[htax_tx_driver_c] 800
[short_packet_test] 1
[short_packet_vsequence] 2
Simulation complete via $finish(1) at time 219270 NS + 45
/opt/coe/cadence/XCELIUM/tools/methodology/UVM/CDNS-1.1d/sv/src/base/uvm_root.svh:457 $finish;
xcelium> exit
```